

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	English-only emotion detection	Excludes non-English speaking	High
2	No user authentication or	Cannot track individual student	High
3	Gemini free tier limit (1500	App unusable after daily quota	High
4	BiLSTM + BERT weights not	Full ML pipeline unavailable on	Medium
5	Single user text input only	Cannot analyze audio, images,	Medium
6	Static keyword dictionary	New slang or domain-specific	Low
7	CSV-based logging	Not scalable beyond ~100K interactions	Low

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
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S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	10 concurrent Streamlit Cloud users	Migrate to FastAPI backend + React frontend on cloud VMs (AWS/GCP)
2	Data Storage	Flat CSV file	Migrate to PostgreSQL or Firebase Realtime DB for structured, indexed storage
3	Performance	Gemini API latency ~2-3s	Cache Gemini responses for identical inputs using Redis
4	Model Deployment	Gemini-only on cloud	Deploy BiLSTM + BERT as REST microservices using FastAPI + Docker on a VPS
5	Security	Gemini key in Streamlit Secrets	Use OAuth2 for user auth; store keys in AWS Secrets Manager

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
6	Multilingual Support	English only	Integrate Google Translate API for preprocessing; retrain on multilingual dataset
7	Analytics	Static Plotly charts	Build a real-time dashboard with WebSocket streaming (Plotly Dash / Grafana)

Future Feature Roadmap

Priority	Feature	Description	ETA
High	User Login & Profiles	Track individual student	Q3 2026
High	Multilingual Support	Detect emotions from Hindi,	Q3 2026
Medium	Voice Input	Analyze emotion from audio recordings via speech-to-text +	

NLP

Medium	Teacher Dashboard	Aggregated class-level emotion	Q4 2026
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Q4 2026

Priority	Feature	Description	ETA
Medium	Mobile App	React Native app wrapping the	Q4 2026
Low	Facial Expression Fusion	Multimodal detection combining	2027
Low	LMS Integration	Plugin for Moodle / Google	2027
Low	Custom Model Fine-Tuning	Let institutions train on their own	2027
		student data	

Scalability Architecture (Future State)

Student (Web / Mobile)

Load Balancer (Nginx)

FastAPI Backend
(Docker Containers)

BiLSTM Service
BERT Service
Gemini Client

PostgreSQL DB
Redis Cache
AWS S3 (Models)